THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which of these is the greatest? 1

(a) 0.45 **(b)** $\frac{1}{2}$ **(c)** $\frac{2}{5}$ **(d)** 0.405

2. If $x = 6$, the value of $2x^2 + 5$ is: 1

(a) 79 **(b)** 78 **(c)** 77 **(d)** 76

3. The number of seconds in 30 days is: 1

(a) 2,59,200 **(b)** 25,88,400 **(c)** 25,92,000 **(d)** 25,93,000

4. The value of $250 \div (5 \times (7 - 2)) + 8 \times (6 + 3)$ is: 1

(a) 82 **(b)** 2139 **(c)** 399 **(d)** 84

5. $1.5 \times 0.2 \times 0.6 =$ 1

(a) 0.18 **(b)** 1.8 **(c)** 0.018 **(d)** 18

6. If $a = 4$ and $b = 5$, the value of $3a^2 - 7b$ is: 1

(a) 13 **(b)** 15 **(c)** 14 **(d)** 12

7. The number of zeros in 1,25,000 × 80 is: 1

(a) 9 **(b)** 8 **(c)** 6 **(d)** 7

8. The value of 'twice the sum of 12 and 8, decreased by 15' is: 1

(a) 10 **(b)** 55 **(c)** 40 **(d)** 25

9. How many times does 0.25 go into 6? 1

(a) 25 **(b)** 23 **(c)** 26 **(d)** 24

10. Simplify $7(x + 3) - 4(x - 2)$: 1

(a) $3x + 13$ **(b)** $3x + 29$ **(c)** $3x - 13$ **(d)** $11x + 29$

11. 1 crore = 1,00,00,000; the number of crores in 4,50,00,000 is: 1

(a) 45 **(b)** 450 **(c)** 0.45 **(d)** 4.5

12. Compare 20 × 13 and 19 × 14. The first minus the second is: 1

(a) -4 **(b)** -5 **(c)** -6 **(d)** -7

13. **Assertion (A):** 2.50 = 2.5. 1
Reason (R): 0.1 = 1/10.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The value of 2x + 3 when x = 4 is 11. 1
Reason (R): In 2x, the number 2 and the letter x are multiplied.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** 45,00,000 is written as 4,500,000 in the international system. 1
Reason (R): In the international system, commas are placed after the first three digits and then after every two digits.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the decimal number exactly halfway between 0.3 and 0.4. 2

17. Find the value of $2x^2 - 3x + 4$ when $x = 5$. 2

18. How many seconds are there in 30 days? Write it in the Indian and the international system. 2

19. Evaluate $250 \div (5 \times (7 - 2)) + 8 \times (6 + 3)$. 2

20. Find the missing number: $1.5 \times \square = 3.6$. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the value of $1.5 \times 0.6 + 2.4 \div 0.6$. 3

22. A rectangle has length $(7x + 3)$ cm and breadth $(2x - 1)$ cm. Find its perimeter as an expression and its value for $x = 5$. 3

23. Arrange 90,87,654, 90,78,654, 98,07,654 in descending order and find the difference between the largest and the smallest. 3

24. Evaluate 14×18 and $(14 + 3) \times (18 - 3)$ and find the difference between the two values. 3

OR

Two friends start from the same place and walk in opposite directions at 4.2 km/h and 3.8 km/h. How far apart are they after 1.5 hours? After how many hours will they be 32 km apart? 3

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25. Write the sum of 4 consecutive numbers starting at n as an expression in n, and evaluate it for $n = 5$. 3

OR

The speed of light is about 3,00,000 km per second. How far does it travel in 60 seconds? Write the answer in words. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Case study. Radha buys 2.5 m of cloth at ₹85.50 per metre. 5

(i) Find the cost of the cloth.

(ii) She pays ₹250. How much change does she get?

(iii) What would 4 m of this cloth cost?

(iv) Her mother has ₹513. How many metres of this cloth can she buy?

27. A tank of capacity 84 litres is filled by a pipe at 9.5 litres per minute, but a leak drains 2.5 litres per minute. Find the net gain per minute and the time to fill the empty tank. 5

OR

In a pattern, figure n has $5n + 2$ tiles. Find 5

(i) the tiles in figure 12,

(ii) the total tiles in figures 1 and 2, and

(iii) how many more tiles figure 12 has than figure 11.

END OF QUESTION PAPER Total: 50 Marks

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